

UGAIN Black Belt – Module 3

Worked exam solutions, explained step by step

This walks through every question with the underlying statistics refreshed from scratch: what each tool is, why we use it here, and every intermediate calculation. Dutch terms from the exam are kept in brackets so the answers map back to your paper.

Note on Q2: the exam refers to an Excel file with the actual machine measurements, which wasn't included. The full method is explained and worked with clearly-labelled *example* sample variances — replace them with the real values from your Excel.

Question 1 — True or false, with justification [/2.5]

The skill being tested here is reading what a statistical result actually says versus what people *assume* it says. Five statements, each judged **correct** / **not correct**.

a) "A hypothesis test shows an expensive process change has a significant positive effect on quality, so it's obvious to invest." → **NOT correct**

The trap is the word *significant*. In statistics, **statistical significance** only means: the observed effect is unlikely to be pure chance (the *p*-value is below your threshold, e.g. 5%). It says **nothing about the size of the effect**, and nothing about money.

Two separate ideas:

- **Statistical significance** – is the effect real (not noise)?
- **Practical / economic significance** – is the effect big enough to be worth the cost?

With a large enough sample, even a microscopic improvement becomes "significant", because the test becomes very sensitive. A change that lifts quality by 0.01% can be statistically significant yet nowhere near worth an expensive retrofit. The investment decision needs an **effect size + cost-benefit analysis**, not just a *p*-value. So the conclusion doesn't follow.

b) "A 95% confidence interval for the fraction defective is [1.4%, 2.2%]. So the process is not suitable to deliver under 1% defectives." → **Correct**

A **confidence interval** (*betrouwbaarheidsinterval*) is a range of plausible values for the unknown true value, built so that 95% of such intervals (over many repetitions) would contain it. Here the plausible range for the true defect rate is **entirely above 1%** — even the most optimistic end (1.4%) is above 1%.

So we have strong evidence the true rate exceeds 1%, which means the process as it stands **cannot** meet a "< 1% defective" requirement. The statement is justified precisely *because the whole interval sits above the 1% line*.

c) "The population mean of X can in theory be estimated to any prescribed accuracy." → **Correct**

The width of a confidence interval for a mean is governed by the **standard error**:

$$SE = \frac{\sigma}{\sqrt{n}}$$

The half-width of the interval is roughly $z \cdot \frac{\sigma}{\sqrt{n}}$. As the sample size n grows, $\sqrt{n}$ grows, so the interval **shrinks toward zero**. There's no theoretical floor: pick any target precision, and there exists an n large enough to reach it. *In practice* you're limited by time and money, but the statement says "in theory", so it's correct.

d) "Cluster sampling will never show non-response bias." → **NOT correct**

These are two unrelated things:

- **Cluster sampling** is a *selection* method: split the population into groups (clusters), randomly pick whole clusters, measure everyone inside.
- **Non-response bias** happens *after* selection: some chosen units don't respond, and the non-responders differ systematically from responders.

No sampling design immunises you against people refusing to answer or units being unmeasurable. Cluster sampling can absolutely suffer non-response bias — if, say, an entire cluster is hard to reach. The claim is false.

e) "To test a hypothesis at 1% significance, you compute a 99% confidence interval." → **Correct**

There's a clean duality between the two: a **two-sided** test at significance level α corresponds exactly to a $(1 - \alpha)$ **confidence interval**.

$$\alpha = 1\% = 0.01 \quad \Rightarrow \quad 1 - \alpha = 0.99 = 99\%$$

The rule: if the hypothesised value falls *outside* the 99% interval, you reject at 1%. So a 1% test $\leftrightarrow$ 99% interval. (*Small nuance: this exact match is for two-sided tests; a one-sided 1% test pairs with a one-sided 99% bound. The statement is correct in the standard framing.*)

Summary: a) $\times$ · b) $\checkmark$ · c) $\checkmark$ · d) $\times$ · e) $\checkmark$

Question 2 — Comparing two machine variances (F-test) [/3]

Situation. Two machines make parts whose critical dimension X should be 10.000. Both are centred correctly, both vary *normally*. We suspect **M1 is more precise than M2** — i.e. M1 has *smaller* spread. We measure $n_1 = 10$ parts from M1 and $n_2 = 15$ from M2, and want a

one-sided 95% confidence interval for the ratio of variances σ_2^2/σ_1^2 , then decide if the suspicion holds.

Step 0 — What does "more precise" mean statistically?

Precision = consistency = *small spread*. Spread is measured by the **variance** σ^2 (the average squared distance from the mean) or its square root the **standard deviation** σ . So:

$$\text{"M1 is more precise than M2"} \iff \sigma_1^2 < \sigma_2^2 \iff \frac{\sigma_2^2}{\sigma_1^2} > 1.$$

Our whole job is to check whether $\frac{\sigma_2^2}{\sigma_1^2}$ is convincingly **greater than 1**.

Step 1 — The sample variances

From the Excel data you compute each **sample variance** (Excel: `=VAR.S(range)`):

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

The $\frac{1}{n-1}$ (not $\frac{1}{n}$) is **Bessel's correction**: it makes s^2 an unbiased estimate of the true σ^2 . Call them s_1^2 (from M1's 10 parts) and s_2^2 (from M2's 15 parts).

Step 2 — Why an F-distribution

When you take the ratio of two independent sample variances drawn from normal populations, that ratio follows an **F-distribution**. The F-distribution is *the* tool for comparing two spreads. It has two "degrees of freedom" parameters — one for each variance — and degrees of freedom here are $n - 1$:

$$\text{numerator df} = n_1 - 1 = 9, \quad \text{denominator df} = n_2 - 1 = 14.$$

The exam hint gives the exact pivot quantity (a quantity whose distribution is known and free of unknowns):

$$\frac{s_1^2}{s_2^2} \cdot \frac{\sigma_2^2}{\sigma_1^2} \sim F(9, 14)$$

Step 3 — Turn the F-statement into a bound on the ratio

We want a *lower* bound for $\rho = \frac{\sigma_2^2}{\sigma_1^2}$ (we care whether $\rho > 1$). Start from the fact that 95% of the $F(9,14)$ distribution lies *above* its 5th percentile $F_{0.05}(9, 14)$:

$$P\left(\frac{s_1^2}{s_2^2} \cdot \rho \geq F_{0.05}(9, 14)\right) = 0.95$$

Rearrange to isolate ρ (multiply both sides by s_2^2/s_1^2):

$$P\left(\rho \geq \frac{s_2^2}{s_1^2} \cdot F_{0.05}(9, 14)\right) = 0.95$$

So the **one-sided 95% confidence interval** is:

$$\boxed{\frac{\sigma_2^2}{\sigma_1^2} \geq \frac{s_2^2}{s_1^2} \cdot F_{0.05}(9, 14)} \quad \text{i.e. } [L, \infty) \text{ with } L = \frac{s_2^2}{s_1^2} F_{0.05}(9, 14).$$

Step 4 — The Excel value

`=F.INV(0.05, 9, 14)` returns the value below which 5% of the $F(9,14)$ distribution falls:

$$F_{0.05}(9, 14) = 0.3305$$

(Sanity check: `F.INV` is the left-tailed inverse, so this is the 5th percentile — a number below 1, exactly what a lower-tail F value should be.)

Step 5 — Worked example (replace with your Excel numbers)

Suppose the Excel gives, say, $s_1^2 = 0.0040$ and $s_2^2 = 0.0150$ (units mm^2). Then:

$$L = \frac{s_2^2}{s_1^2} \cdot F_{0.05}(9, 14) = \frac{0.0150}{0.0040} \times 0.3305 = 3.75 \times 0.3305 = 1.24$$

So the 95% interval would be $\frac{\sigma_2^2}{\sigma_1^2} \in [1.24, \infty)$.

Step 6 — Decision

If the lower bound $L > 1$, the entire interval lies above 1. We're 95% confident $\sigma_2^2 > \sigma_1^2$, so **M2 really is less precise than M1 — the suspicion is justified.**

****If $L \leq 1$ **** (the interval includes 1), we *cannot* conclude M1 is more precise; the observed difference could be sampling luck.

In the example above $L = 1.24 > 1$, so the suspicion would be confirmed. Plug your real s_1^2, s_2^2 into Step 5 and apply this rule.

Question 3 — Process capability [/2]

Situation. Customer wants steel-sheet width in [1400, 1460] mm. The process produces widths that are **normal** with mean $\mu = 1440$ mm and standard deviation $\sigma = 10$ mm. So Lower Spec Limit LSL = 1400, Upper Spec Limit USL = 1460.

a) Capability of the process

Two indices. C_p asks "is the process spread narrow enough to fit inside the spec?" — it ignores centring:

$$C_p = \frac{USL - LSL}{6\sigma} = \frac{1460 - 1400}{6 \times 10} = \frac{60}{60} = 1.0$$

(The 6σ is the natural process width: $\pm 3\sigma$ on each side covers 99.73% of a normal distribution.)

But $C_p = 1.0$ assumes the process is *centred*. It isn't: the spec centre is $(1400 + 1460)/2 = 1430$, while $\mu = 1440$ — shifted 10 mm high. **C_{pk}** accounts for this by taking the *worst* of the two sides:

$$\begin{aligned} C_{pk} &= \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right) = \min\left(\frac{1460 - 1440}{30}, \frac{1440 - 1400}{30}\right) \\ &= \min\left(\frac{20}{30}, \frac{40}{30}\right) = \min(0.667, 1.333) = \boxed{0.67} \end{aligned}$$

Reading it: $C_p = 1.0$ says the spread *could* just fit, but $C_{pk} = 0.67$ says the off-centre mean wrecks it. Anything below 1.33 is generally considered not capable; 0.67 is poor. The bottleneck is the **upper** side (the process runs too high).

b) Expected % scrap (uitval)

Scrap = parts outside [1400, 1460]. We convert each spec limit to a **z-score** (how many standard deviations from the mean):

$$z = \frac{\text{value} - \mu}{\sigma}$$

Upper tail (above 1460): $z = \frac{1460 - 1440}{10} = +2.0$. From the normal table, $P(Z > 2) = 1 - 0.9772 = 0.0228 = \mathbf{2.28\%}$.

Lower tail (below 1400): $z = \frac{1400 - 1440}{10} = -4.0$. $P(Z < -4) \approx 0.00003 = 0.003\%$ — negligible.

$$\text{Total scrap} \approx 2.28\% + 0.003\% \approx \boxed{2.28\%}$$

Essentially all the scrap comes from exceeding the upper limit — exactly what the low C_{pk} warned about.

c) Two ways to improve capability

1. **Re-centre the process** — shift the mean from 1440 down to the spec centre 1430. This rebalances the tails. With $\mu = 1430$ both limits sit at $z = \pm 3$, and scrap drops from ~2.3% to ~0.27%. (Cheapest fix: just an adjustment, no new variation removed.)
2. **Reduce the variation** — lower σ (better machine setup, more uniform raw material, tighter control). Smaller σ raises both C_p and C_{pk} and shrinks both tails.

(A third, usually unavailable, option is widening the tolerance — only if the customer allows it.)

d) Max % scrap tolerated to meet the 6-sigma criterion

The Six Sigma quality target, allowing for the conventional long-term **1.5 σ drift**, is **3.4 defects per million opportunities (DPMO)**:

$$\frac{3.4}{1,000,000} = 0.00034\%$$

$$\boxed{\text{at most } 3.4 \text{ ppm} = 0.00034\%}$$

(For comparison, this process currently runs at ~22,800 ppm — about 6,700 $\times$ too many defects, roughly a 3.5 σ level.)

Question 4 — Control-chart theory [/1.5]

a) Why may we assume normality and work with means / standard deviations?

Because of the **Central Limit Theorem (CLT)**. A control chart doesn't plot individual measurements — it plots **subgroup averages** $\bar{x}$. The CLT says that the distribution of an *average* tends toward a **normal distribution** as the subgroup size grows, *even if the individual measurements are not normal*. So the plotted averages are approximately normal, which justifies normal-based control limits at $\mu \pm 3\sigma$ (covering 99.73% of in-control variation).

b) Which causes of variation can a control chart detect?

Special / assignable causes (bijzondere / aanwijsbare oorzaken). Every process has two

kinds of variation:

- **Common cause** – the inherent, random background noise; points dance randomly *within* the limits.
- **Special cause** – something out of the ordinary (tool breaks, wrong material, setup error); it produces a *signal*: a point outside the limits, a run, or a trend.

The chart's job is to flag **special causes** so you investigate them, while leaving common-cause noise alone (over-reacting to noise is "tampering" and makes things worse).

c) When do we use the range as the best estimator?

When **subgroups are small** (rule of thumb $n \lesssim 8-10$). The **range** $R = x_{\max} - x_{\min}$ is trivial to compute and, for small subgroups, almost as efficient an estimate of spread as the standard deviation s . For *larger* subgroups the range throws away information (it only uses two points), so the **s-chart** becomes the better choice.

Question 5 — Bias vs variance from confusion matrices [/4]

Background: reading a confusion matrix and computing accuracy

A **confusion matrix** cross-tabulates actual class (rows) against predicted class (columns). The diagonal = correct predictions. **Accuracy** = (correct) / (total):

$$\text{Accuracy} = \frac{\text{correct predictions}}{\text{total predictions}}$$

Computing all six (training has 1000 rows, test has 600):

Model	Training accuracy	Test accuracy	Gap (train – test)
A	$(480+485)/1000 = \mathbf{96.5\%}$	$(180+190)/600 = \mathbf{61.7\%}$	~35 pts
B	$(380+360)/1000 = \mathbf{74.0\%}$	$(190+180)/600 = \mathbf{61.7\%}$	~12 pts
C	$(420+430)/1000 = \mathbf{85.0\%}$	$(200+215)/600 = \mathbf{69.2\%}$	~16 pts

a) Which model is high variance, high bias, optimal?

The **bias-variance trade-off** is read from two numbers: how well the model fits the training data, and how big the drop-off is on unseen test data.

- **High variance (overfitting)**: memorises the training set (near-perfect train score) but fails to generalise (big train→test drop). → **Model A**: 96.5% train, but collapses to 61.7% on test — a ~35-point gap. *Classic overfitting*.

- **High bias (underfitting):** too simple to capture the pattern, so it's weak even on the training data, and the gap to test is small. → **Model B:** lowest training score of all (74%) and only 61.7% on test. It just can't fit. *Classic underfitting.*
- **Optimal:** good training fit and the best generalisation. → **Model C:** 85% train, **69.2% test (the highest test accuracy)**, with a moderate, healthy gap. *Best balance.*

Answer: A = high variance · B = high bias · C = optimal. The decisive tell: A fits training almost perfectly but generalises worst; B can't even fit training; C generalises best.

b) High bias in a decision tree — what to change?

High bias = the tree is **too simple / underfitting**. Make it **more flexible** so it can capture more structure:

- **Increase the maximum depth** (let it grow more levels).
- **Lower the minimum samples per split / per leaf** (allow more, finer splits).
- **Reduce pruning** (less aggressive cost-complexity pruning).
- **Add or engineer more informative features** for it to split on.

All of these let the tree fit the training pattern better, lowering bias. (The opposite — limiting depth, pruning more — is what you'd do for the *high-variance* Model A.)

Question 6 — Distribution families and Var[C] [/4]

Situation. Capacitors ("elco's"), nominal 800 μF , non-conform if capacity < 720 μF . Sold in lots of 100. On average **2 in 40 are non-conform** → $p = 2/40 = 0.05$. Mean capacity is actually **820 μF** (not 800). **175.3 orders/week, 3.6 lots/order.**

a) Which distribution family fits each variable?

Quick refresher on each family and its mean:

- **Poisson** - counts of independent events in a fixed interval. Mean = the rate.
- **Exponential** - continuous *waiting time between* such events. Mean = 1/rate.
- **Normal** - a continuous quantity scattering symmetrically around a centre.
- **Bernoulli** - a single yes/no trial. Mean = probability of "1".
- **Binomial** - number of "successes" in n independent yes/no trials. Mean = np .
- **Uniform** - every value in a range equally likely (a distractor here — nothing fits it).

	normaal	Bernoulli	exponentieel	uniform	Poisson	binomiaal
B (orders/week)					$E[B] = 175.3$	
T (time between orders)			$E[T] = 1/175.3 \text{ wk}$ $\approx 0.96 \text{ h}$			
C (capacity, μF)		$E[C] = 820$				
X (conform yes/no)		$E[X] = 0.95$				
D (non-conform per lot)						$E[D] = 5$

Reasoning for each:

- **B** counts orders in a week (a fixed interval) → **Poisson**, mean 175.3.
- **T** is the gap between consecutive Poisson arrivals → **Exponential**, mean $1/175.3$ week ≈ 0.96 hour.
- **C** is a continuous measured quantity fluctuating around a centre → **Normal**, mean 820.
- **X** is one part's conform/not outcome → **Bernoulli**. $E[X] = P(\text{conform}) = 1 - 0.05 = 0.95$. *(If you instead code X as "non-conform = 1", then $E[X] = 0.05$ — still Bernoulli.)*
- **D** is the count of non-conform parts among 100 independent parts → **Binomial** (100, 0.05), mean $np = 100 \times 0.05 = 5$.

b) For which variable is the i.i.d. assumption unrealistic, and why?

"i.i.d." = elements that are **independent** and **identically distributed**. Several variables strain this — multiple answers are accepted:

- **B (orders/week)**: weeks are **not identically distributed** — seasonality, holidays, promotions, end-of-quarter pushes all shift the ordering rate. Weeks aren't interchangeable.
- **D (non-conform per lot)** (*and similarly C*): parts made consecutively on one line share process conditions (tool wear, drift, a bad batch of material). Defects therefore **cluster** rather than occurring independently — when the process drifts, you get a run of bad parts. So the "100 independent draws with constant p " assumption behind the binomial is shaky; defects are positively autocorrelated.

Briefly: **B** because weeks differ systematically over time (not identically distributed);
D/C because parts off the same line are not independent (common process causes

correlate them).

c) Compute $\text{Var}[C]$

We know two things about C (Normal): its mean $\mu = 820$, and that **5% of parts fall below 720 μF** (that's what "non-conform" means, at rate 0.05). Use the second fact to back out σ .

Step 1 — z-score of the 720 cut-off. Standardise:

$$z = \frac{720 - 820}{\sigma} = \frac{-100}{\sigma}$$

Step 2 — this z must be the 5th percentile. Since $P(C < 720) = 0.05$, the standardised cut-off equals the standard-normal 5th percentile:

$$z_{0.05} = -1.645$$

Step 3 — solve for σ .

$$\frac{-100}{\sigma} = -1.645 \quad \Rightarrow \quad \sigma = \frac{100}{1.645} = 60.80 \mu\text{F}$$

Step 4 — variance is σ^2 .

$$\boxed{\text{Var}[C] = \sigma^2 = (60.80)^2 \approx 3696 \mu\text{F}^2}$$

The whole trick: a single tail probability (5% below 720) plus the known mean pins down the standard deviation, because in a normal distribution percentile $\leftrightarrow$ z-score $\leftrightarrow$ (value – mean)/ σ .

Question 7 — Regression line and causal interventions [/3]

This question separates **seeing** (observing a correlation) from **doing** (intervening) — the heart of causal inference. T = annealing temperature ($^{\circ}\text{C}$), S = product strength (MPa).

a) Sketch the regression line $f(x) = E[S \mid T = x]$

$E[S \mid T = x]$ is the **conditional mean** of strength given temperature — the *average* S at each value of T . On the scatter plot, draw the line (or smooth curve) that threads through the **vertical centre of the point cloud** at every temperature: for each x , the height of the line is the mean of the S -values sitting above that x . This is the ordinary best-fit / least-squares line through the data.

b) Causal diagram $S \rightarrow T$, intervene $\text{do}(T = 730)$. Sketch ~10 dots.

Here **S causes T** — strength is the *cause*, temperature is the *effect*. The observed S-T correlation exists *because S drives T*.

Now we **intervene** and force $T = 730$ (the "do" operator: $\text{do}(T = 730)$). Crucially, **we are fixing the effect, not the cause**. Forcing an effect tells you nothing about its cause and doesn't change it — you can't alter strength by overriding the thermometer. So:

- *S* keeps its **original, full marginal distribution** (unchanged).
- *T* is pinned at 730.

Draw: dots in a **vertical line at $T = 730$** , spread across the **entire natural range of S** (the full top-to-bottom spread S shows in the original data). The original correlation is irrelevant — intervening on the effect cuts the link, so S is unconstrained.

c) Causal diagram $T \rightarrow S$, intervene $\text{do}(T = 730)$. Sketch ~10 crosses (second figure).

Now **T causes S** — temperature is the *cause*, strength is the *effect*. Forcing $T = 730$ now **propagates through the causal mechanism** to S. So *S* follows the **conditional distribution $P(S \mid T = 730)$** .

Draw: crosses in a **vertical line at $T = 730$** , but **clustered tightly around the regression line's predicted value $E[S \mid T = 730]$** , scattered only by the small residual (conditional) spread at that temperature.

Why b) and c) look completely different — the key insight

Same data, same intervention temperature, opposite pictures:

	b) $S \rightarrow T$ (intervene on the effect)	c) $T \rightarrow S$ (intervene on the cause)
Does forcing T change S?	No	Yes
S after intervention	full marginal spread of S	conditional spread around $E[S \mid 730]$
Picture at $T = 730$	dots spanning all of S	crosses hugging the regression line

This is the difference between **conditioning** (passively observing parts that happened to be at 730°C) and **intervening** (actively setting 730°C). The same observational scatter is consistent with *either* causal direction — which is exactly why you can't read causation off a correlation, and why the causal diagram matters before you act.

One-line recap of the toolkit used

- Q1 - significance $\neq$ practical importance; CI $\leftrightarrow$ test duality; $SE = \sigma / \sqrt{n}$.
- Q2 - F-distribution for comparing two variances; one-sided CI via **F.INV**.
- Q3 - C_p , C_{pk} , z-scores $\rightarrow$ tail probabilities; 3.4 DPMO.
- Q4 - Central Limit Theorem; special vs common cause; range for small subgroups.
- Q5 - confusion-matrix accuracy; bias-variance trade-off; tuning tree complexity.
- Q6 - distribution families & their means; i.i.d. realism; recovering σ from a tail probability.
- Q7 - regression as conditional mean; the do-operator; intervening on a cause vs an effect.